IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE/EIE PART II: MEng, BEng and ACGI

Corrected Copy

SIGNALS AND LINEAR SYSTEMS

Tuesday, 11 June 10:00 am

Time allowed: 2:00 hours

Question 1 d(ii) the number 1 should not be next to the diagram See Page 2

There are THREE questions on this paper.

Answer ALL questions.

Question One carries 40% of the marks. The other 2 questions each carry 30%.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.T. Stathaki
Second Marker(s) : P.L. Dragotti

© Imperial College London

1. This question carries 40% of the mark.

(a) Consider each of the following continuous-time signals. For each case, specify if the signal is causal/non-causal, periodic/non-periodic, odd/even. If the signal is periodic specify its period.

(i) $x(t) = 2\cos(2\pi t)$ [2]

(ii) $x(t) = 5e^{2\text{sign}(t)\cdot t}$ [2]

(b) Consider the signal

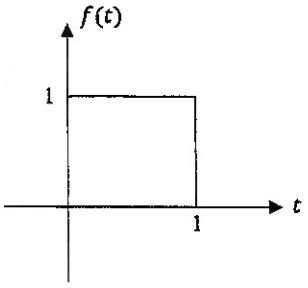
$$x(t) = \begin{cases} 1-t, & 0 \le t \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Sketch each of the following signals.

(i) $x\left(-\frac{t}{3}+1\right)$ [2]

(ii) $x(2t+5)$ [2]

(c) By applying the property of convolution, find the system's output (i.e., zero-state response) given that the impulse response is $h(t) = e^{-t}u(t)$ and that the input $f(t)$ is as shown in Figure 1(c) below. We denote with $u(t)$ the continuous-time unit step function.

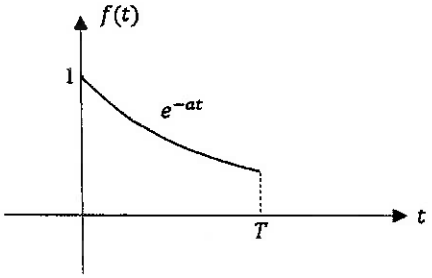


[6]

Figure 1(c)

(d) Derive the Fourier transform of the signals shown in Figures 1(d)(i)-(ii) below. The parameters a and b shown in Figures 1(d)(i)-(ii) below are constant parameters.

(i)

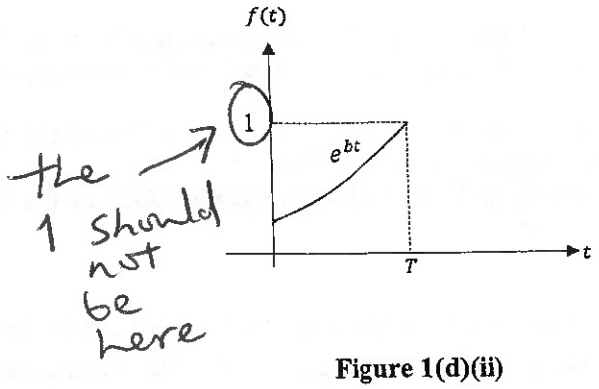


[3]

Figure 1(d)(i)

 © Imperial College London

(ii)



the 1 should not be here

[3]

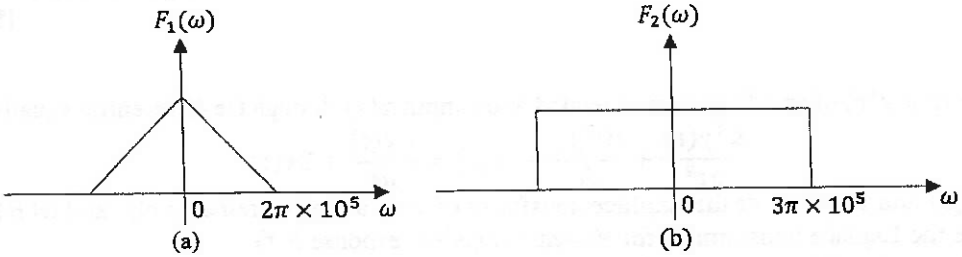
Figure 1(d)(ii)

(e) The output $y(t)$ of a continuous-time LTI system is related to the input $x(t)$ through the differential equation:

$$\frac{dy(t)}{dt} + 2y(t) = 2x(t)$$

Determine the frequency response of the system and sketch the asymptotic behavior of its Bode plots. **[5]**

(f) Figures (a) and (b) below show the Fourier spectra of signals $f_1(t)$ and $f_2(t)$. Determine the Nyquist sampling rates for $f_1(t)$ and $f_2(t)$. (Hint: Use the width property of the convolution.)



(i) $f_1(t)$ **[2]**
(ii) $f_2(t)$ **[2]**
(iii) $f_1^2(t)$ **[2]**
(iv) $f_2^3(t)$ **[2]**
(v) $f_1(t)f_2(t)$ **[2]**

(g) Consider the discrete-time, causal LTI system with input $x[n]$ and output $y[n]$ related through the difference equation:

$$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n] - \frac{1}{2}x[n-1]$$

(i) Find the analytical expression and the Region of Convergence (ROC) of the z −transform of the impulse response of the above system.
[Hint: Use the fact that the z-transform $\frac{z}{z-a}$ corresponds to the function $a^n u[n]$ if $|z| > |a|$ and the function $-a^n u[-n-1]$ if $|z| < |a|$. The function $u[n]$ is the discrete-time unit step function.] **[3]**

(ii) Find the analytical expression and the Region of Convergence (ROC) of the z −transform of the output if $x[n] = \left(\frac{1}{2}\right)^n u[n]$. **[2]**

 © Imperial College London

2. This question carries 30% of the mark.

(a) (i) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time signal $x(t) = e^{-at}u(t)$, with $u(t)$ the continuous-time unit step function. [3]

(ii) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the continuous-time signal $x(t) = -e^{-at}u(-t)$. [3]

(iii) Find the analytical expression and the Region of Convergence (ROC) of the Laplace transform of the Dirac function $\delta(t)$. [3]

(b) Consider a continuous-time causal signal $x(t)$, ($x(t) = 0$, $t < 0$) with Laplace transform $X(s) = \frac{s+2}{s-2}$. Find the analytical expression of $x(t)$, using the results obtained in question (a) above and also find the region of convergence of $X(s)$.
[Hint: you may write the above Laplace transform as $X(s) = 1 + \frac{4}{s-2}$]. [4]

(c) Consider a continuous Linear Time-Invariant (LTI) system with input the signal $x(t) = \delta(t)$ (the Dirac function) and output the signal $y(t) = -\frac{2}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t)$.

(i) Determine the transfer function $H(s)$ of the system and its Region of Convergence (ROC). [5]

(ii) Determine the Laplace transform of the output of the system and its Region of Convergence (ROC) when the input is the signal $x(t) = u(t)$ (the continuous-time unit step function). The ROC of $Y(s)$ is the intersection of the ROCs of $X(s)$ and $H(s)$. [5]

(d) The output $y(t)$ of an LTI system is related to the input $x(t)$ through the differential equation:

$$\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} - 2y(t) = \frac{dx(t)}{dt} + 2x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of the system's impulse response $h(t)$.

(i) Determine $H(s)$ as a ratio of two polynomials. [3]

(ii) Determine $h(t)$ for each of the following cases:
1. The system is stable.
2. The system is causal.

[4]

© Imperial College London

3. This question carries 30% of the mark.

(a) Signals $f_1(t) = \Pi(10^4 t)$ and $f_2(t) = \frac{1}{2}(\delta(t - 0.5 \cdot 10^{-4}) + \delta(t + 0.5 \cdot 10^{-4}))$ are applied at the inputs of ideal lowpass filters with frequency responses $H_1(\omega) = 10^4 \Pi\left(\frac{\omega}{40000\pi}\right)$ and $H_2(\omega) = \Pi\left(\frac{\omega}{20000\pi}\right)$ respectively, with

$$\Pi(t) = \begin{cases} 1 & |t| < 0.5 \\ 0.5 & |t| = 0.5 \\ 0 & \text{otherwise} \end{cases}$$

The outputs $y_1(t)$ and $y_2(t)$ of these filters are multiplied to obtain the signal $y(t) = y_1(t)y_2(t)$ as shown in Figure 3a below.

(i) Find and sketch $F_1(\omega)$ and $F_2(\omega)$. **[4]**
(ii) Sketch $H_1(\omega)$ and $H_2(\omega)$. **[2]**
(iii) Find and sketch $Y_1(\omega)$ and $Y_2(\omega)$. **[2]**
(iv) Find the Nyquist sampling rate of $y_1(t)$, $y_2(t)$ and $y(t)$. **[2]**
(For any function $x(t)$ we denote its Fourier transform with $X(\omega)$).

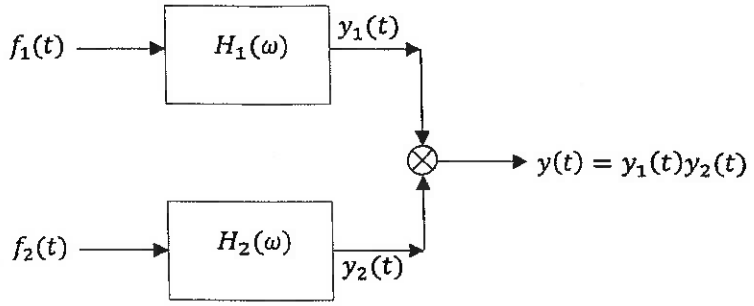


Figure 3a

(b) (i) Consider a discrete signal $x[n]$ with z −transform $X(z)$. Find the z −transforms of the signals $x[n+3]$ and $x[-n+1]$ as functions of the z-transform $X(z)$. **[4]**

(ii) Consider a signal $y[n]$ which is related to two signals $x_1[n]$ and $x_2[n]$ by

$$y[n] = x_1[n+3] * x_2[-n+1]$$

where

$$x_1[n] = \left(\frac{1}{2}\right)^n u[n] \text{ and } x_2[n] = \left(\frac{1}{3}\right)^n u[n]$$

with $u[n]$ the discrete unit step function. Use the results obtained in question (b)(i) above to determine the z-transform of $y[n]$. **[4]**

(c) Consider an even sequence $x[n]$ (i.e., $x[n] = x[-n]$) with z −transform $X(z)$.

(i) From the definition of the z −transform, show that $X(z) = X(1/z)$. **[3]**
(ii) From your result in part (c)(i), show that if a pole (zero) of $X(z)$ occurs at $z = z_0$, then a pole (zero) must also occur at $z = 1/z_0$. **[3]**

Verify the results in part (c)(ii) for each of the following sequences:

(iii) $\delta[n+1] + \delta[n-1]$ **[3]**
(iv) $\delta[n+1] - \frac{5}{2}\delta[n] + \delta[n-1]$ with $\delta[n]$ the discrete impulse function. **[3]**

 © Imperial College London